MIL-T-3235B(OS)
5 March 1976
SUPERSEDING
MIL-T-3235A(Wep)
12 May 1961

MILITARY SPECIFICATION

TUBE, COPPER, SEAMLESS (FOR TORPEDO USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers seamless copper tube for use in torpedoes.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issues in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

STANDARDS

Federal

FED-STD-151	Metals; Test Methods
FED-STD-185	Identification Marking of Copper and Copper Base Alloy Mill Products

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 4710

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SPECIFICATIONS

Military

*MIL-C-3993 Copper and Copper-Base Alloy Mill Products; Packaging of

(Copies of specifications, standards, and publications required by contractors in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The tube shall be cold drawn to size from any of the following: Copper 102, 103, 120, or 122 (types OF, OFXLP, DLP, or DHP, respectively).

3.2 Chemical composition. The chemical composition shall conform to the requirements shown in table I.

TABLE I. Chemical composition (percent).

Copper No.	Designation	Copper (incl. silver)	Phosphorus
102	OF	99.95 min.	-
103	OFXLP	99.95 min.[1]	0.001 - 0.005
120	DLP	99.90 min.	0.004 - 0.012
122	DHP	99.9 min.	0.015 - 0.040

[1] The value includes copper + silver + phosphorus.

3.3 Hardness. Unless otherwise specified in the contract or order, the tube shall be furnished in (hard) drawn temper. Hardness requirements shall conform to table II.

TABLE II. Hardness requirements.

Temper	Wall thickness (inches)	Rockwell hardness 30T scale
Light drawn	All	30 - 60
Hard drawn	All	55 minimum

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3.4 Tolerances.

3.4.1 Wall thickness. Wall thickness of tube shall not vary by more than the amounts shown in table III. When specified in the contract or order that the tolerances shall be all plus or minus, the tolerances shown in table III should be doubled.

TABLE III. Tolerances in wall thickness, plus or minus, inches.

Wall thickness (inch)	Outside diameter					
	1/32 to 1/8, inclusive	Over 1/8 to 5/8, inclusive	Over 5/8 to 1, inclusive	Over 1 to 2, inclusive	Over 2 to 4, inclusive	Over 4 to 7, inclusive
Under 0.018	0.002	0.001	0.0015	0.002	-	-
0.018 incl to 0.025	0.003	0.002	0.002	0.0025	-	-
0.025 incl to 0.035	0.003	0.0025	0.0025	0.003	0.004	-
0.035 incl to 0.058	0.003	0.003	0.0035	0.0035	0.005	0.007
0.058 incl to 0.083	-	0.0035	0.004	0.004	0.005	0.008
0.083 incl to 0.120	-	0.004	0.005	0.005	0.007	0.009
0.120 incl to 0.165	-	0.005	0.006	0.006	0.008	0.010
0.165 incl to 0.220	-	0.007	0.0075	0.008	0.010	0.012
0.165 incl to 0.220	-	0.007	0.0075	0.008	0.010	0.012
0.220 incl to 0.284	-	-	0.009	0.010	0.012	0.014
0.284 incl to 0.380	-	-	0.11	0.012	0.014	0.016

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3.4.2 Diameters. The average outside or inside diameter of any tube, determined by several measurements at a given cross section, shall conform to the specified diameter within the tolerances shown in table IV. When specified in the contract or order that the tolerances shall be all plus or minus, the tolerances shown in table IV should be doubled.

TABLE IV. Tolerances in average diameter[1]

Specified diameter (inches)	Tolerance plus or minus (inch)
1/8 and under	0.002
Over 1/8 to 5/8 inch	0.002
Over 5/8 to 1 inch	0.0025
Over 1 to 2, inclusive	0.003
Over 2 to 3, inclusive	0.004
Over 3 to 4, inclusive	0.005
Over 4 to 5, inclusive	0.006

[1] Tolerance applies to inside or outside diameter.

3.4.3 Length.

3.4.3.1 Random straight lengths. Unless otherwise specified in the contract or order, the tube shall be furnished in random lengths of not less than 12 nor more than 16 feet.

3.4.3.2 Straight lengths. When exact lengths are specified, in the contract or order, the tolerances shall be as given in table V. Tolerances are all plus; if all minus tolerances are desired, use the same values. If tolerances plus or minus are desired, halve the values given.

3.4.3.3 Squareness of cut. All tubes, up to and including 5/8 inch outside diameter, must not depart from the square by more than 0.010 inch and by not more than 0.016 inch per inch in outside diameters of more than 5/8 inch.

3.5 Hot and cold bend. Material of the tube shall not crack on the outside of the bend when subjected to hot and cold bend tests as specified in 4.4.4 and 4.4.5.

3.6 Embrittlement. When specified in the contract or order, the tube shall be capable of passing the embrittlement test specified in 4.4.3.

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*3.7 Identification marking. When specified in the contract or order (see 6.2), identification marking shall be in accordance with FED-STD-185. Tubes shall be free of indented or raised markings.

TABLE V. Tolerances for copper, tube, straight lengths.

Length	Tolerances,[1] (inch)		
	For outside diameters up to 1 inch inclusive	For outside diameters over 1 to 4 inches inclusive	For outside diameters over 4 inches
Specific lengths:			
Up to 6 inches, inclusive	1/32	1/16	-
Over 6 inches to 2 feet, inclusive	1/16	3/32	1/8
Over 2 to 6 feet, inclusive	3/32	1/8	1/4
Over 6 to 14 feet, inclusive	1/4	1/4	1/4
Over 14 feet	1/2	1/2	1/2
Specific lengths with ends	1	1	1
Stock lengths with or without ends	1[2]	1[2]	1[2]

[1] Applicable only to full length pieces.
[2] As stock lengths are cut and placed in stock in advance of orders, departure from this tolerance is not practical.

3.8 Workmanship. The tube shall be clean, smooth, and free from grooving and indentations, cracks, flaws, scale, pits, or other defects; and it shall be round, straight, of the proper dimensions, and with ends cut square.

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4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Sampling for examination and testing shall be in accordance with FED-STD-151, as specified herein.

*4.1.2 The requirements of 4.2 through 4.5 inclusive shall be applied when it is determined that the manufacturer's method of quality control is not acceptable to the Government.

4.2 Sampling for lot acceptance inspection.

4.2.1 Inspection lot. For the purposes of inspection, a lot shall consist of the following quantities of tubes of the same size and temper offered for delivery at the same time:

a. Tubes with outside diameter above 1/2 inch: 5,000 pounds, or a fraction thereof.

b. Tubes with outside diameter above 1/2 inch and under: 2,000 pounds or a fraction thereof.

*4.2.1.1 One extra tube in each lot ordered in exact length shall be furnished, or one sample tube of nominal length shall be selected for chemical analysis and mechanical tests. A test piece approximately 2 feet long shall be taken from the extra or sample tube.

4.2.2 Sampling for examination.

4.2.2.1 Visual examination. From each inspection lot, a representative sample of tubes shall be selected in accordance with table VI and the procedures of MIL-STD-105 and shall be visually examined as specified in 4.3.1.

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*4.2.2.2 Dimensional examination. From each inspection lot, a representative sample of tubes shall be selected in accordance with table VI and the procedures of MIL-STD-105 and shall be dimensionally examined as specified in 4.3.2. The samples selected for dimensional examination may be the same as those selected for visual examination.

TABLE VI. Sampling for visual and dimensional examination AQL (approximately) = 1.5 percent defective.

Number of tubes in an inspection lot	Number of tubes in sample	Acceptance number (defectives)	Rejection number (defectives)
40 or under	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3
501 to 800	75	3	4
801 to 1300	110	4	5
1301 and over	150	5	6

*4.2.3 Sampling for chemical analysis. Two ounces of clean chips shall be taken from the sample tube selected as specified in 4.2.1.1.

4.2.4 Sampling for embrittlement test. When applicable, one sample tube shall be selected for the embrittlement test specified in 4.4.3.

4.2.5 Sampling for cold bend test. When applicable, a strip not over 1-1/2 inches wide shall be taken from one sample tube for the cold bend test specified in 4.4.4.

4.2.6 Sampling for hot bend test (for tubes under 2-3/8 inches in outside diameter). When applicable, one sample piece of tube in full section and of sufficient length shall be selected for the hot bend test as specified in 4.4.5.

4.2.7 Sampling for hardness test. When applicable, one sample tube shall be selected for the hardness test specified in 4.4.2.

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4.3 Examination.

4.3.1 Visual. Each sample tube selected shall be visually examined to verify conformance to the requirements of this specification.

4.3.2 Dimensional. Each sample tube shall be examined to verify conformance to the dimensional tolerances of this specification.

4.4 Test procedures.

4.4.1 Chemical analysis. The sample selected in accordance with 4.2.3 shall be analyzed to determine conformance to the requirements of 3.2.

4.4.2 Hardness test. The tube selected in accordance with 4.2.7 shall be tested for hardness in accordance with method 243.1 of FED-STD-151.

4.4.3 Embrittlement test. When required, (see 3.6), the tube shall be tested as follows: Samples shall be heated to a temperature of 800° to 875° C for a period of 20 minutes in a furnace containing an atmosphere of hydrogen. After their removal from the furnace, the samples shall be polished, etched if desired, and examined under a microscope at a magnification of not less than 75 diameters. The samples shall not show gassing or open grain structure characteristic of embrittlement.

4.4.4 Cold bend test. After annealing, the sample selected as specified in 4.2.5 shall be bent flat on itself without cracking on the outside of the bend. A crack in the bend test is defined as a break which extends all the way across the outside face of the specimen or extends past mid-thickness of the specimen on one side.

4.4.5 Hot bend test. The sample selected for test as specified in 4.2.6, shall be plugged and filled with sand or other suitable material and heated to a low red heat (1,300° - 1,400° F). While hot, the tube shall withstand bending around a mandrel 1-1/2 times the inside diameter of the tubing through 180° without the development of cracks or flaws on the outside of the bent portion.

4.5 Rejection. If any specimen fails to conform to this specification, the entire lot shall be rejected, subject to the retest provisions in accordance with FED-STD-151.

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5. PACKAGING

*5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.3).

*5.1.1 Levels A and B. The material shall be packed in accordance with MIL-C-3993.

5.1.2 Level C. The product shall be separated by size, composition, and temper and packed in accordance with the manufacturer's standard practice into containers of a type and size commonly used for the purpose, in such a manner as to insure acceptance by carrier for transportation at the lowest rate applicable and to afford maximum protection for normal hazards of transportation.

5.2 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Copper tubes covered by this specification are intended for use in torpedoes.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Temper required (see 3.3).

c. Diameter and wall thickness required (see 3.4.1 and 3.4.2).

d. Whether continuous identification marking is required (see 3.7).

e. Whether material is to be packed by Level A, B, or C (see 5.1).

f. Maximum gross weight of containers.

g. Whether embrittlement test is required (see 3.6).

h. Whether tolerances shall be all plus or all minus (see 3.4.1, 3.4.2, 3.4.3, and 3.4.3.2).

i. Whether straight lengths of less than 12 feet or more than 16 feet are required (see 3.4.3.1).

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6.3 The requirements of packing (see 5.1) and marking for shipment (see 5.2) specified herein apply to direct shipment for Government activities and also, where specified, to contracts or orders between the manufacturer and the Government prime contractor.

6.4 The contractor may be required to replace tubes in which injurious defects are revealed subsequent to acceptance without expense to the Government.

*6.5 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-3235B(OS) Tube, Copper, Seamless (For Torpedo Use)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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